THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Sun	primary source
2	**(d)** radiation	no medium required
3	**(a)** trap air, a poor conductor	air between fibres
4	**(b)** hot air rises	convection
5	**(b)** convection currents	warm water rises
6	**(c)** water cycle	evaporation, condensation, rain
7	**(b)** infiltration	recharges groundwater
8	**(c)** degree Celsius	°C
9	**(b)** a poor conductor of heat	insulator
10	**(b)** A black cloth	dark absorbs

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat transfer from the hotter to the colder part of an object (1); heating a spoon in hot tea (1).	1+1
12	Heat transfer by movement of a fluid (1); boiling water or sea breeze (1).	1+1
13	Transfer of heat without a medium (1); through space as heat waves (1).	1+1
14	Wool traps air which is a poor conductor (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place wax beads along a rod (1); heat one end (1); beads fall in order showing heat flow (1).	1+1+1
16	Drop potassium permanganate crystal in a flask heated from below (1.5); coloured streams rise and fall (1.5).	1½+1½
17	Day: land hotter, hot air rises, cool wind from sea (1.5); night: reverse (1.5).	1½+1½
	OR: Evaporation (1); condensation and clouds (1); precipitation and infiltration (1).	1+1+1
18	Solids/medium contact (1); fluids movement (1); no medium (1).	1+1+1
	OR: Metal conducts heat to food (1.5); plastic insulates (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Conduction, convection, radiation (1). (ii) Explanation (3). (iii) Radiation (1).	1+3+1
	OR: (i) Convection, conduction, radiation (1). (ii) Hot air lighter (1). (iii) Conduction (1). (iv) Radiation (2).	1+1+1+2
20	(i) Evaporation, condensation, precipitation (2). (ii) Provides heat (1). (iii) Infiltration into aquifers (2).	2+1+2
	OR: (i) Sea and land breeze (1). (ii) Land hotter (1). (iii) Sea warmer (1). (iv) Fishermen and sailors use them (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** conduction	through the solid
2	**(d)** Aluminium	metal
3	**(b)** air trapped between them	insulation by air
4	**(a)** from sea to land during the day	land heats faster
5	**(b)** radiation	heat waves
6	**(c)** evaporation	Sun's heat
7	**(d)** aquifers	pore spaces
8	**(b)** 35 °C and 42 °C	body temperature range
9	**(c)** conduction, convection and radiation	vacuum and silvered walls
10	**(a)** the water cycle	continuous process

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They are poor conductors and do not burn hands (2).	2
12	Cool wind from the sea to the land during the day (2).	2
13	They reflect more heat (2).	2
14	Liquid to vapour (1); vapour to liquid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Vacuum prevents conduction/convection (1.5); silvered walls reduce radiation (1.5).	1½+1½
16	Dark colours absorb more heat (3).	3
17	Falling water table (1.5); rainwater harvesting and afforestation (1.5).	1½+1½
	OR: Stores water underground for wells and borewells (3).	3
18	Heated air is less dense and rises by convection (3).	3
	OR: White paint, ventilation, shades (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Heat passes easily (1). (ii) Metals; wood, plastic (2). (iii) Handles, wool, thermos (2).	1+2+2
	OR: (i) Heat water with dye (2). (ii) Flask, burner (1). (iii) Coloured water rises, spreads and sinks (2).	2+1+2
20	(i) Falling water table (1). (ii) Rainwater harvesting (1). (iii) Seepage of water (1). (iv) Check dams or percolation pits (2).	1+1+1+2
	OR: (i) Colour matters (1). (ii) Dark absorbs, light reflects (2). (iii) White in summer, dark in winter (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.